

The Two Faces of Time: Three Quantities Physics Has Conflated

Coordinate time (label) · Proper time τ (geometric) · Duration $E(a)$ (thermodynamic)

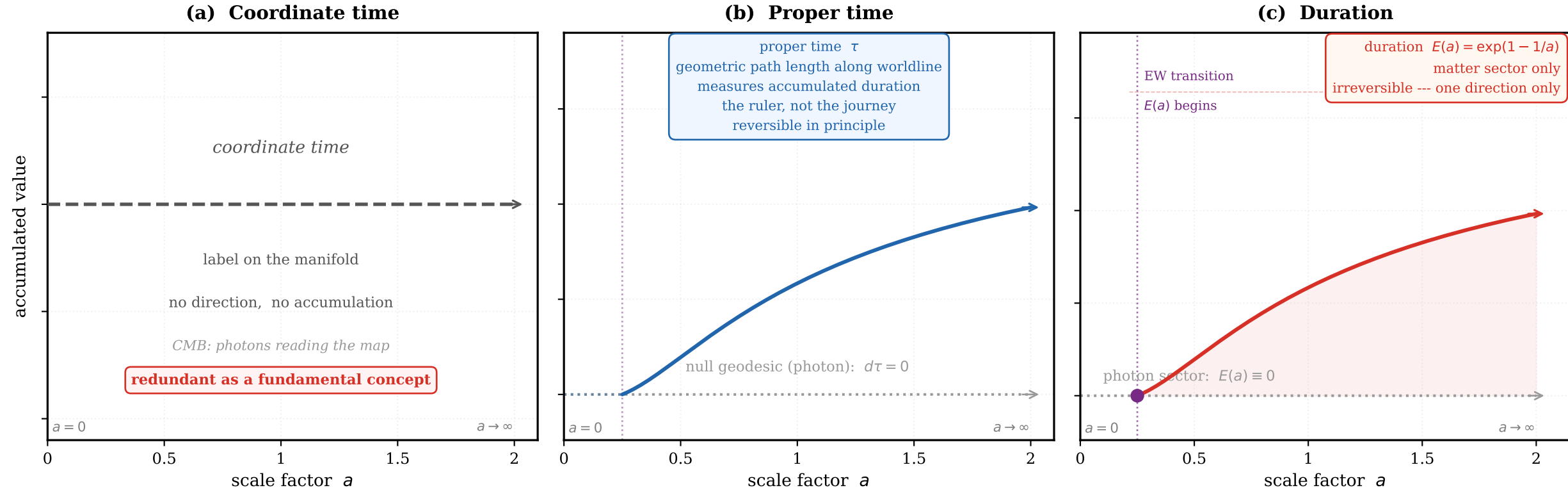


Fig. 1. Three quantities physics has conflated. (a) Coordinate time: a label on the manifold, redundant as a fundamental concept. (b) Proper time τ : geometric path length along a worldline, reversible in principle; photons accumulate none ($d\tau = 0$). (c) Duration $E(a) = \exp(1 - 1/a)$: thermodynamic, irreversible, beginning at the electroweak (EW) transition, accumulating on massive particles only. The matter sector and photon sector are separated by construction. IAM framework: Mahaffey (2026), doi:10.5281/zenodo.18702042